

INTRODUCTION TO GIT & GITHUB



WHY VERSION CONTROL?

Have you ever...

- Ended up with a folder full of files named report_final_3_FINAL_this_version?
- Gone back to a past project and forgotten what you did?
- Had your computer crash and lose your progress completely?
- Made a mistake and wanted to return to an earlier version of your work?
- Wanted to work together with someone on the same project without clashing?

WHAT IS VERSION CONTROL?

Version control: the practice of tracking and managing changes to software code

With version control you can:

- **Track provenance:** Understand what version of the data and code was used to produce specific results.
- **Ensure reproducibility:** Allow others to reproduce findings by providing access to the exact versions of data and code used.
- **Collaborate Effectively:** Facilitate collaboration among researchers by managing changes and tracking contributions.

GIT + GITHUB FOR VERSION CONTROL

Git

- Underlying version control software
- Runs locally on your computer
- Tracks changes in your files over time

GitHub

- Cloud-based hosting service
- Manages your Git repositories online
- Allows for sharing and collaboration

LET'S GIT STARTED!

[UKRN Git & GitHub training](#)

HAPPY GIT AND GITHUB FOR THE USER

Connect Git, GitHub, RStudio

CREATING A REPOSITORY

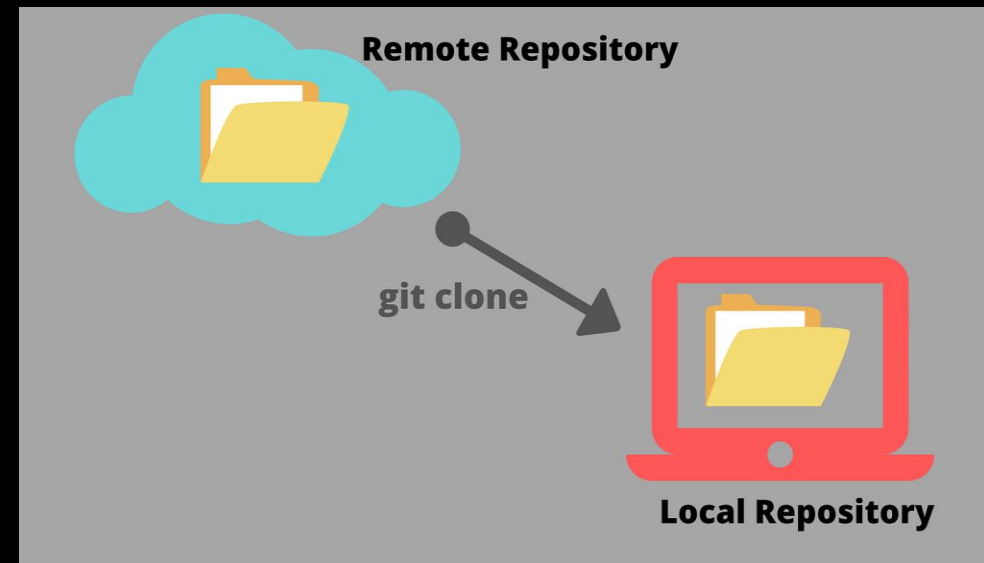
Repository: the storage place where your project's files and history of changes are kept

[Creating a repository](#)

CLONING

Cloning: the process of transferring a GitHub repository from the cloud to your computer (local)

We can use RStudio to help us with this process: <https://github.com/babichmorrowc/bristol-uganda-data-week/tree/main/intro-to-git-github>



COMMITTING, PUSHING, & PULLING

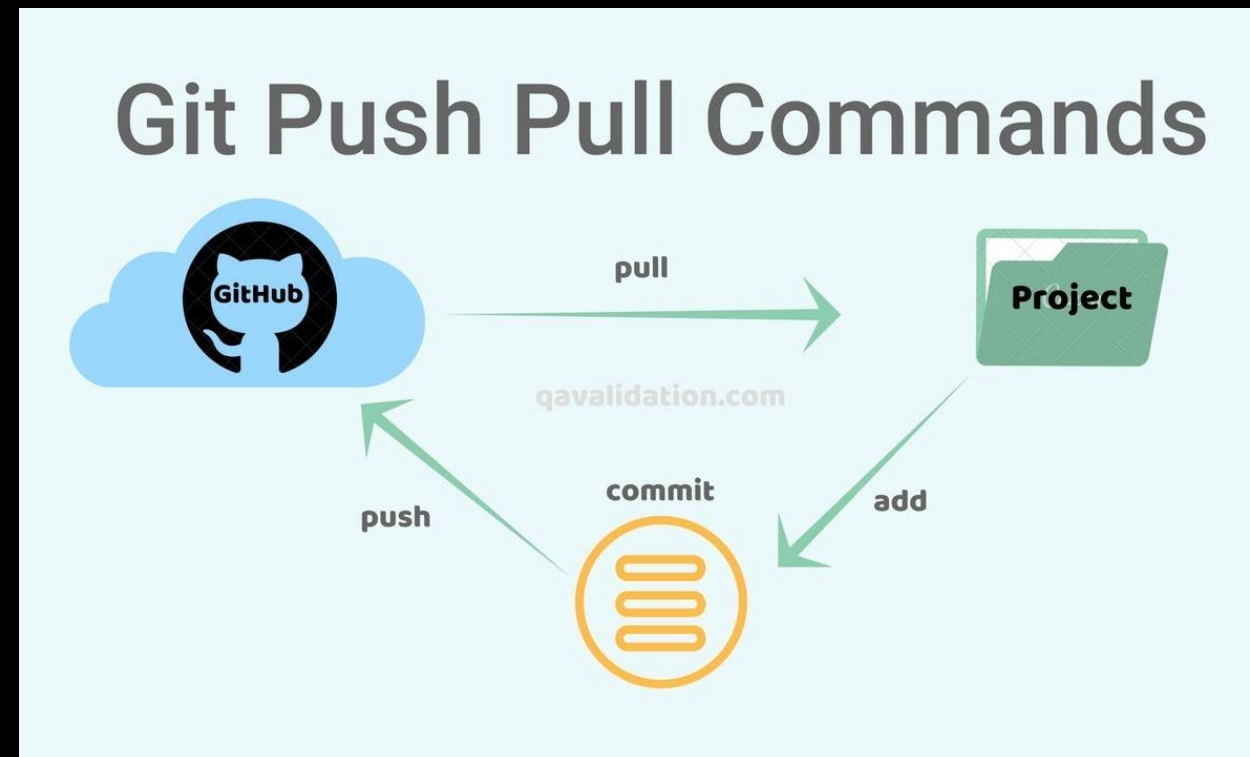
Adding: telling Git to start monitoring the new versions of a file

Committing: telling Git to save a new snapshot version of your file(s)

Pulling: integrating changes from the remote repository into your local repository

Pushing: integrating changes from your local repository into the remote repository

[Committing, pulling, and pushing](#)



BRANCHING

Branching: the process of creating a new version ("branch") of the code from the main version ("tree")

- The branch is still connected to the main version of the code
- It retains all of the history of the tree
- Allows for collaboration

Branching

GIT Branch and its Operations.

